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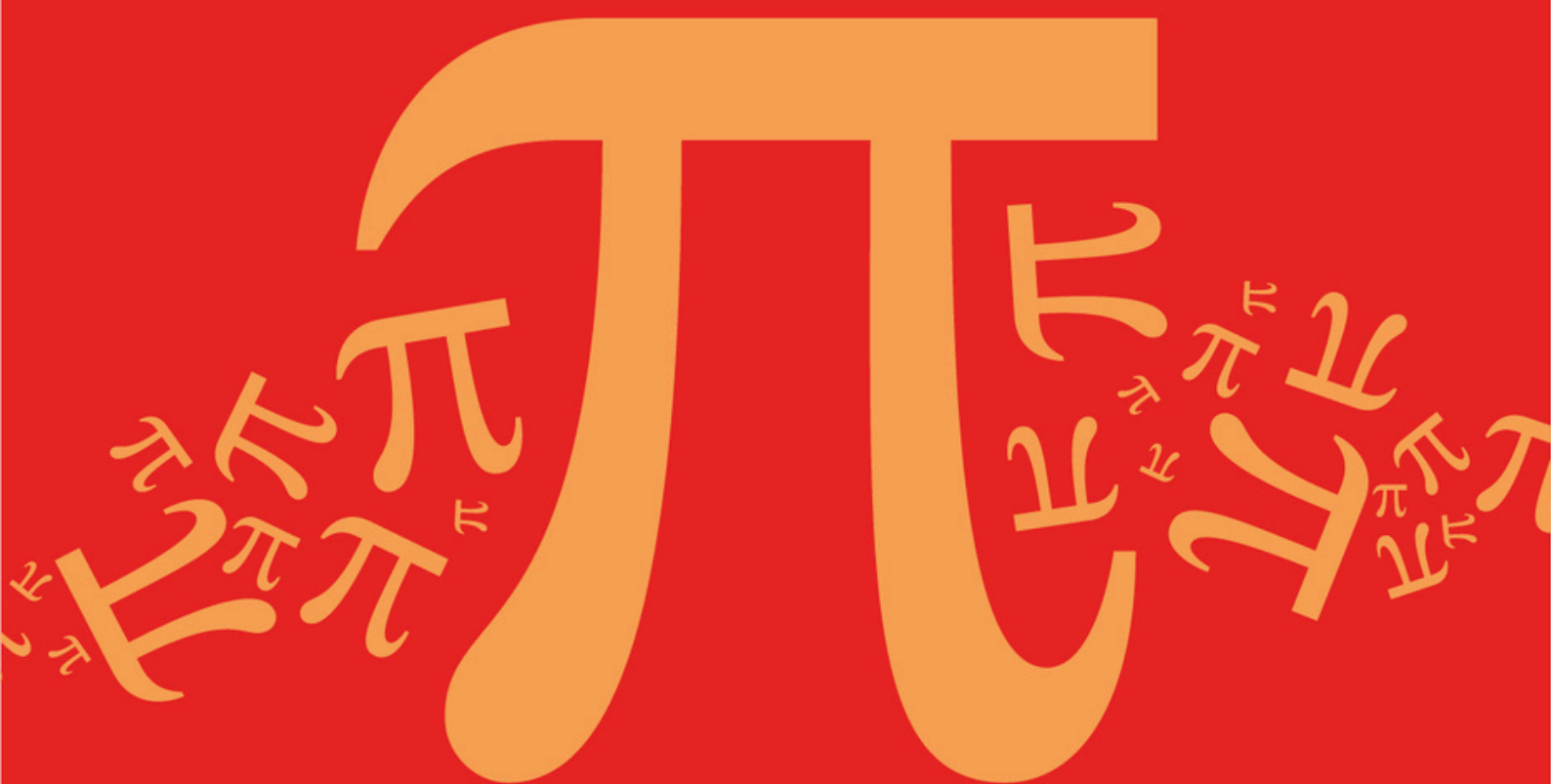
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AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB4: Travelling Salesperson

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

DB: Networks

Sub topic

DB4: Travelling Salesperson

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
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Name : _____

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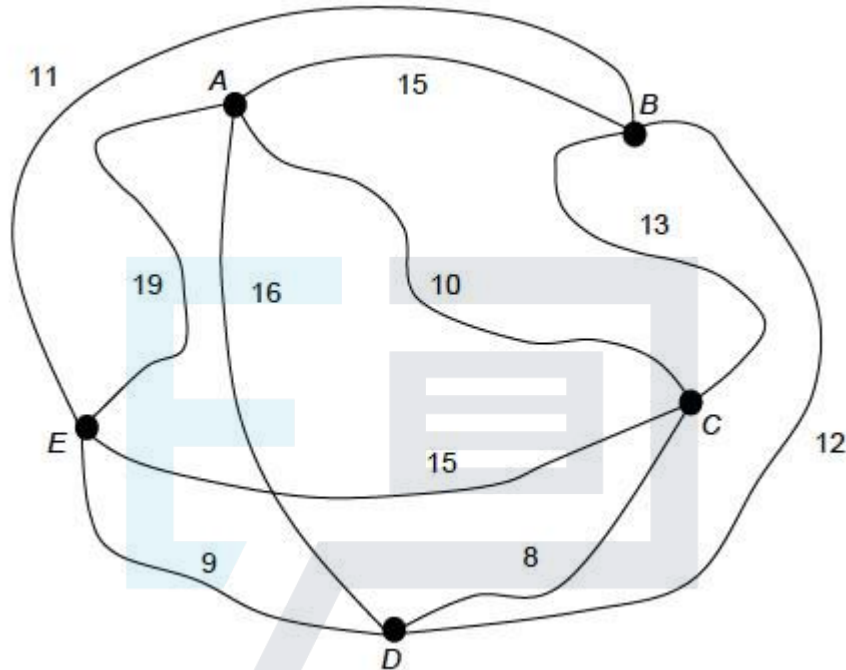
Total Marks Available : _____

Total Marks Received : _____

Q1.

Charlotte is visiting a city and plans to visit its five monuments: *A*, *B*, *C*, *D* and *E*.

The network shows the time, in minutes, that a typical tourist would take to walk between the monuments on a busy weekday morning.



Charlotte intends to walk from one monument to another until she has visited them all, before returning to her starting place.

- (a) Use the nearest neighbour algorithm, starting from *A*, to find an upper bound for the minimum time for Charlotte's tour. (2)
- (b) By deleting vertex *B*, find a lower bound for the minimum time for Charlotte's tour. (3)
- (c) Charlotte wants to complete the tour in 52 minutes. Use your answers to parts **(a)** and **(b)** to comment on whether this could be possible. (1)
- (d) Charlotte takes 58 minutes to complete the tour. Evaluate your answers to part **(a)** and part **(b)** given this information. (1)
- (e) Explain how this model for a typical tourist's tour may not be applicable if the tourist walked between the monuments during the evening. (1)

(Total 8 marks)

Q2.

Tony delivers paper to five offices, *A*, *B*, *C*, *D* and *E*. Tony starts his deliveries at office *E*

and travels to each of the other offices once, before returning to office *E*. Tony wishes to keep his travelling time to a minimum.

The table shows the travelling times, in minutes, between the offices.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>A</i>	–	10	16	20	8
<i>B</i>	10	–	21	15	9
<i>C</i>	16	21	–	10	23
<i>D</i>	20	15	10	–	17
<i>E</i>	8	9	23	17	–

- (a) Find the travelling time of the tour *ACDBEA*. (1)
- (b) Hence write down a tour, starting at *E*, which has the same total travelling time as your answer to part (a). (1)
- (c) Use the nearest neighbour algorithm, starting at *E*, to find an upper bound for the minimum travelling time for Tony's tour. (4)
- (d) By deleting *E*, find a lower bound for the minimum travelling time for Tony's tour. (4)
- (e) Sketch a network showing the edges that give the lower bound in part (d), and comment on its significance. (2)

(Total 12 marks)

Q3.

Sarah is a mobile hairdresser based at *A*. Her day's appointments are at five places: *B*, *C*, *D*, *E* and *F*. She can arrange the appointments in any order. She intends to travel from one place to the next until she has visited all of the places, starting and finishing at *A*. The following table shows the times, in minutes, that it takes to travel between the six places.

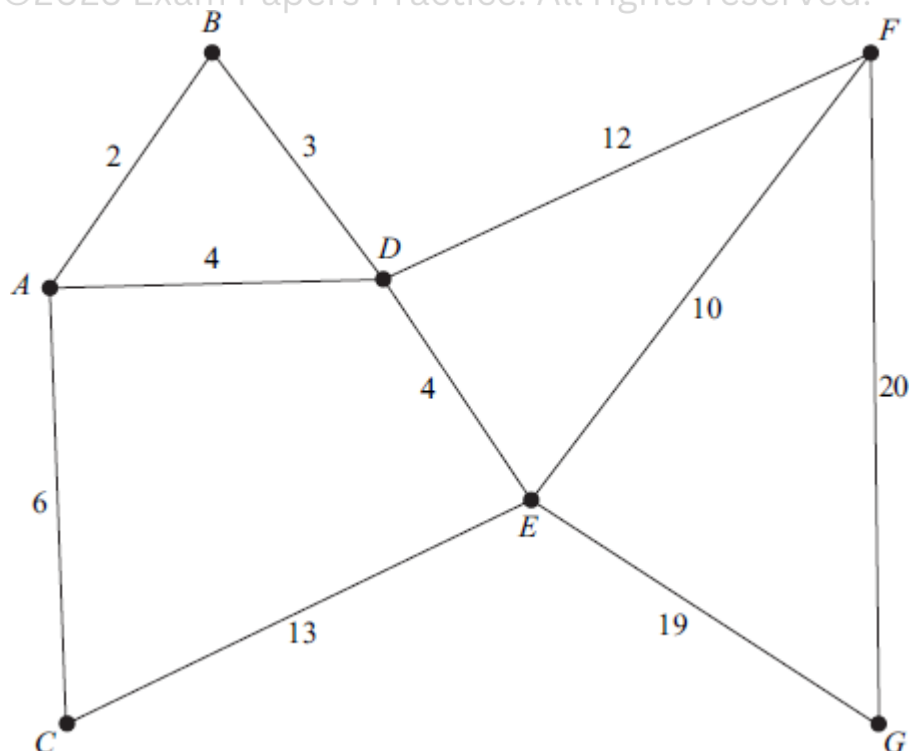
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	–	15	11	14	27	12
<i>B</i>	15	–	13	19	24	15
<i>C</i>	11	13	–	10	19	12

<i>D</i>	14	19	10	–	26	15
<i>E</i>	27	24	19	26	–	27
<i>F</i>	12	15	12	15	27	–

- (a) Sarah decides to visit the places in the order *ABCDEF**A*. Find the travelling time of this tour. (1)
- (b) Explain why this answer can be considered as being an upper bound for the minimum travelling time of Sarah's tour. (2)
- (c) Use the nearest neighbour algorithm, starting from *A*, to find another upper bound for the minimum travelling time of Sarah's tour. (4)
- (d) By deleting *A*, find a lower bound for the minimum travelling time of Sarah's tour. (4)
- (e) Sarah thinks that she can reduce her travelling time to 75 minutes. Explain why she is wrong. (1)
- (Total 12 marks)

Q4.

The diagram shows the locations of some schools. The number on each edge shows the distance, in miles, between pairs of schools.



Sam, an adviser, intends to travel from one school to the next until he has visited all of the schools, before returning to his starting school. The shortest distances for Sam to travel between some of the schools are shown in **Table 1** below.

(a) Complete **Table 1**.

Table 1

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
<i>A</i>	–	2	6	4		16	27
<i>B</i>	2	–	8	3		15	26
<i>C</i>	6	8	–	10		22	32
<i>D</i>	4	3	10	–		12	23
<i>E</i>					–		
<i>F</i>	16	15	22	12		–	20
<i>G</i>	27	26	32	23		20	–

(2)

(b) (i) On the completed **Table 1**, use the nearest neighbour algorithm, starting from *B*, to find an upper bound for the length of Sam's tour.

(4)

(ii) Write down Sam's actual route if he were to follow the tour corresponding to the answer in part (b)(i).

(2)

(iii) Using the nearest neighbour algorithm, starting from each of the other vertices in turn, the following upper bounds for the length of Sam's tour were obtained: 77, 77, 77, 76, 77 and 76.

Write down the best upper bound.

(1)

(c) (i) On **Table 2** below, showing the order in which you select the edges, use Prim's algorithm, starting from *A*, to find a minimum spanning tree for the schools *A*, *B*, *C*, *D*, *F* and *G*.

Table 2

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>	<i>G</i>
<i>A</i>	–	2	6	4	16	27
<i>B</i>	2	–	8	3	15	26

C	6	8	–	10	22	32
D	4	3	10	–	12	23
F	16	15	22	12	–	20
G	27	26	32	23	20	–

(4)

(ii) Hence find a lower bound for the length of Sam's minimum tour.

(3)

(iii) By deleting each of the other vertices in turn, the following lower bounds for the length of a minimum tour were found: 50, 48, 52, 51, 56 and 64.

Write down the best lower bound.

(1)

(d) Given that the length of a minimum tour is T miles, use your answers to parts (b) and (c) to write down the smallest interval within which you know T must lie.

(2)

(Total 19 marks)

Q5.

Rupta, a sales representative, has to visit six shops, A , B , C , D , E and F . Rupta starts at shop A and travels to each of the other shops once, before returning to shop A . Rupta wishes to keep her travelling time to a minimum.

The table shows the travelling times, in minutes, between the shops.

	A	B	C	D	E	F
A	–	16	10	25	26	40
B	16	–	20	19	18	50
C	10	20	–	14	22	31
D	25	19	14	–	11	32
E	26	18	22	11	–	42
F	40	50	31	32	42	–

(a) Find the travelling time of the tour $ACFDEBA$.

(1)

(b) Use the nearest neighbour algorithm, starting at A , to find an upper bound for the

travelling time for Rupta's tour.

(4)

- (c) By deleting A , find a lower bound for the travelling time for Rupta's tour.

(4)

- (d) Sketch a network showing the edges that give you the lower bound in part (c) and comment on its significance.

(2)

(Total 11 marks)

Q6.

Fred delivers bread to five shops, A , B , C , D and E . Fred starts his deliveries at shop B , and travels to each of the other shops once before returning to shop B . Fred wishes to keep his travelling time to a minimum.

The table shows the travelling times, in minutes, between the shops.

	A	B	C	D	E
A	-	3	11	15	5
B	3	-	18	12	4
C	11	18	-	5	16
D	15	12	5	-	10
E	5	4	16	10	-

- (a) Find the travelling time for the tour $BACDEB$.

(1)

- (b) Use the nearest neighbour algorithm, starting at B , to find another upper bound for the travelling time for Fred's tour.

(3)

- (c) By deleting C , find a lower bound for the travelling time for Fred's tour.

(4)

- (d) Sketch a network showing the edges that give you the lower bound in part (c) and comment on its significance.

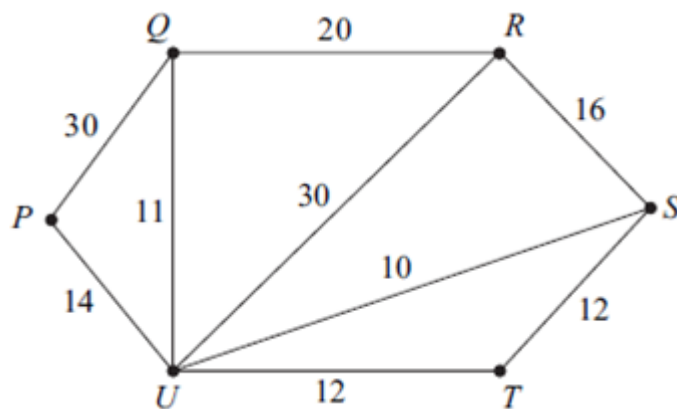
(2)

(Total 10 marks)

Q7.

Mrs Jones is a spy who has to visit six locations, P , Q , R , S , T and U , to collect information. She starts at location Q , and travels to each of the other locations, before returning to Q . She wishes to keep her travelling time to a minimum.

The diagram represents roads connecting different locations. The number on each edge represents the travelling time, in minutes, along that road.



- (a) (i) Explain why the shortest time to travel from P to R is 40 minutes. (2)
- (ii) Complete the table below, in which the entries are the shortest travelling times, in minutes, between pairs of locations.

	P	Q	R	S	T	U
P	–	25				
Q	25	–	20	21	23	11
R		20	–			
S		21		–		
T		23			–	12
U		11			12	–

- (b) (i) Use the nearest neighbour algorithm on the table, starting at Q , to find an upper bound for the minimum travelling time for Mrs Jones's tour. (2)
- (ii) Mrs Jones decides to follow the route given by the nearest neighbour algorithm. (4)
- Write down her route, showing all the locations through which she passes. (2)
- (c) By deleting Q from the table, find a lower bound for the travelling time for Mrs Jones's tour. (5)

(Total 15 marks)

Q8.

There is a one-way system in Manchester. Mia is parked at her base, B , in Manchester and intends to visit four other places, A , C , D and E , before returning to her base. The

following table shows the distances, in kilometres, for Mia to drive between the five places A , B , C , D and E . Mia wants to keep the total distance that she drives to a minimum.

To From	A	B	C	D	E
A	–	1.7	1.9	1.8	2.1
B	3.1	–	2.5	1.8	3.7
C	3.1	2.9	–	2.7	4.2
D	2.0	2.8	2.1	–	2.3
E	2.2	3.6	1.9	1.7	–

- (a) Find the length of the tour $BECDAB$. (1)
- (b) Find the length of the tour obtained by using the nearest neighbour algorithm starting from B . (4)
- (c) Write down which of your answers to parts (a) and (b) would be the better upper bound for the total distance that Mia drives. (1)
- (d) On a particular day, the council decides to reverse the one-way system. For this day, find the length of the tour obtained by using the nearest neighbour algorithm starting from B . (4)

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(Total 10 marks)

Q9.

Phil, a squash coach, wishes to buy some equipment for his club. In a town centre, there are six shops, G , I , N , R , S and T , that sell the equipment.

The time, in seconds, to walk between each pair of shops is shown in the table.

Phil intends to check prices by visiting each of the six shops before returning to his starting point.

	G	I	N	R	S	T
G	–	81	82	86	72	76
I	81	–	80	82	68	73
N	82	80	–	84	70	74
R	86	82	84	–	74	70
S	72	68	70	74	–	64
T	76	73	74	70	64	–

- (a) Use the nearest neighbour algorithm starting from S to find an upper bound for

Phil's minimum walking time.

(4)

- (b) Write down a tour starting from N which has a total walking time equal to your answer to part (a).

(1)

- (c) By deleting S , find a lower bound for Phil's minimum walking time.

(5)

(Total 10 marks)

Q10.

Liam is taking part in a treasure hunt. There are five clues to be solved and they are at the points A , B , C , D and E . The table shows the distances between pairs of points. All of the distances are functions of x , **where x is an integer**.

Liam must travel to all five points, starting and finishing at A .

	A	B	C	D	E
A	–	$x + 6$	$2x - 4$	$3x - 7$	$4x - 14$
B	$x + 6$	–	$3x - 7$	$3x - 9$	$x + 9$
C	$2x - 4$	$3x - 7$	–	$2x - 1$	$x + 8$
D	$3x - 7$	$3x - 9$	$2x - 1$	–	$2x - 2$
E	$4x - 14$	$x + 9$	$x + 8$	$2x - 2$	–

- (a) The nearest point to A is C .

- (i) By considering AC and AB , show that $x < 10$.

(2)

- (ii) Find two other inequalities in x .

(2)

- (b) The nearest neighbour algorithm, starting from A , gives a **unique** minimum tour $ACDEBA$.

- (i) By considering the fact that Liam's tour visits D immediately after C , find two further inequalities in x .

(3)

- (ii) Find the value of the integer x .

(4)

- (iii) Hence find the total distance travelled by Liam if he uses this tour.

(2)

(Total 13 marks)

Q11.

Angelo is visiting six famous places in Palermo: *A*, *B*, *C*, *D*, *E* and *F*. He intends to travel from one place to the next until he has visited all of the places before returning to his starting place. Due to the traffic system, the time taken to travel between two places may be different dependent on the direction travelled.

The table shows the times, in minutes, taken to travel between the six places.

From \ To	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	–	25	20	20	27	25
<i>B</i>	15	–	10	11	15	30
<i>C</i>	5	30	–	15	20	19
<i>D</i>	20	25	15	–	25	10
<i>E</i>	10	20	7	15	–	15
<i>F</i>	25	35	29	20	30	–

- (a) Give an example of a Hamiltonian cycle in this context.

(2)

- (b) (i) Show that, if the nearest neighbour algorithm starting from *F* is used, the total travelling time for Angelo would be 95 minutes.

(3)

- (ii) Explain why your answer to part (b)(i) is an upper bound for the minimum travelling time for Angelo.

(2)

- (c) Angelo starts from *F* and visits *E* next. He also visits *B* before he visits *D*. Find an improved upper bound for Angelo's total travelling time.

(3)

(Total 10 marks)

Q12.

- (a) James is solving a travelling salesperson problem.

- (i) He finds the following upper bounds: 43, 40, 43, 41, 55, 43, 43.

Write down the best upper bound.

(1)

- (ii) James finds the following lower bounds: 33, 40, 33, 38, 33, 38, 38.

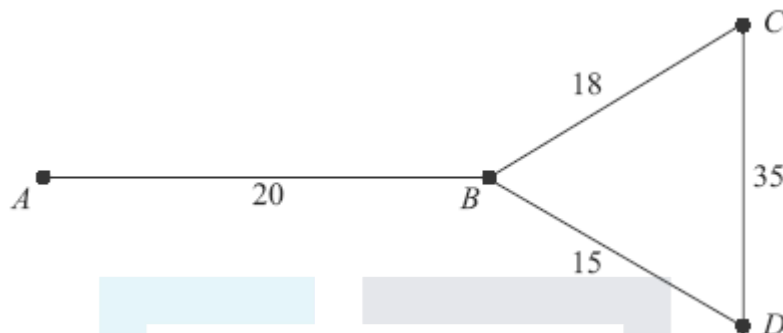
Write down the best lower bound.

(1)

- (b) Karen is solving a different travelling salesperson problem and finds an upper bound of 55 and a lower bound of 45. Write down an interpretation of these results.

(1)

- (c) The diagram below shows roads connecting 4 towns, A , B , C and D . The numbers on the edges represent the lengths of the roads, in kilometres, between adjacent towns.



Xiong lives at town A and is to visit each of the other three towns before returning to town A . She wishes to find a route that will minimise her travelling distance.

- (i) Complete the figure below to show the shortest distances, in kilometres, between **all** pairs of towns.

	A	B	C	D
A	—		38	
B		—		
C	38		—	
D				—

(2)

- (ii) Use the nearest neighbour algorithm on the figure above to find an upper bound for the minimum length of a tour of this network that starts and finishes at A .

(3)

- (iii) Hence find the actual route that Xiong would take in order to achieve a tour of the same length as that found in part (c)(ii).

(2)

(Total 10 marks)

Q13.

David, a tourist, wishes to visit five places in Rome: Basilica (B), Coliseum (C), Pantheon

(*P*), Trevi Fountain (*T*) and Vatican (*V*). He is to start his tour at one of the places, visit each of the other places, before returning to his starting place.

The table shows the times, in minutes, to travel between these places. David wishes to keep his travelling time to a minimum.

	<i>B</i>	<i>C</i>	<i>P</i>	<i>T</i>	<i>V</i>
<i>B</i>	–	43	57	52	18
<i>C</i>	43	–	18	13	56
<i>P</i>	57	18	–	8	48
<i>T</i>	52	13	8	–	51
<i>V</i>	18	56	48	51	–

- (a) (i) Find the total travelling time for the tour *TPVBCT*. (1)
- (ii) Find the total travelling time for David's tour using the nearest neighbour algorithm starting from *T*. (4)
- (iii) Explain why your answer to part (a)(ii) is an upper bound for David's minimum total travelling time. (2)
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- (b) (i) By deleting *B*, find a lower bound for the total travelling time for the minimum tour. (5)
- (ii) Explain why your answer to part (b)(i) is a lower bound for David's minimum total travelling time. (2)
- (c) Sketch a network showing the edges that give the lower bound found in part (b)(i) and comment on its significance. (2)

(Total 16 marks)

Q14.

Mark is driving around the one-way system in Leicester. The following table shows the times, in minutes, for Mark to drive between four places: *A*, *B*, *C* and *D*. Mark decides to start from *A*, drive to the other three places and then return to *A*.

Mark wants to keep his driving time to a minimum.

From \ To	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	–	8	6	11
<i>B</i>	14	–	13	25
<i>C</i>	14	9	–	17
<i>D</i>	26	10	18	–

- (a) Find the length of the tour *ABCD A*. (2)
- (b) Find the length of the tour *ADCBA*. (1)
- (c) Find the length of the tour using the nearest neighbour algorithm starting from *A*. (4)
- (d) Write down which of your answers to parts (a), (b) and (c) gives the best upper bound for Mark's driving time. (1)
- (Total 8 marks)**

Q15.

- (a) Mark is staying at the Grand Hotel (*G*) in Oslo. He is going to visit four famous places in Oslo: Aker Brygge (*A*), the National Theatre (*N*), Parliament House (*P*) and the Royal Palace (*R*).

The figures in the table represent the walking times, in seconds, between the places.

	Grand Hotel (<i>G</i>)	Aker Brygge (<i>A</i>)	National Theatre (<i>N</i>)	Parliament House (<i>P</i>)	Royal Palace (<i>R</i>)
Grand Hotel (<i>G</i>)	–	165	185	65	160
Aker Brygge (<i>A</i>)	165	–	155	115	275
National Theatre (<i>N</i>)	185	155	–	205	125
Parliament House (<i>P</i>)	65	115	205	–	225
Royal Palace (<i>R</i>)	160	275	125	225	–

Mark is to start his tour from the Grand Hotel, visiting each place once before

returning to the Grand Hotel. Mark wishes to keep his walking time to a minimum.

- (i) Use the nearest neighbour algorithm, starting from the Grand Hotel, to find an upper bound for the walking time for Mark's tour. (4)
- (ii) By deleting the Grand Hotel, find a lower bound for the walking time for Mark's tour. (5)
- (iii) The walking time for an optimal tour is T seconds. Use your answers to parts (a)(i) and (a)(ii) to write down a conclusion about T . (1)

- (b) Mark then intends to start from the Grand Hotel (G), visit three museums, Ibsen (I), Munch (M) and Viking (V), and return to the Grand Hotel. He uses public transport. The table shows the minimum travelling times, in minutes, between the places.

From \ To	Grand Hotel (G)	Ibsen (I)	Munch (M)	Viking (V)
Grand Hotel (G)	–	20	17	30
Ibsen (I)	15	–	32	16
Munch (M)	26	18	–	21
Viking (V)	19	27	24	–

- (i) Find the length of the tour $GIMVG$. (1)
- (ii) Find the length of the tour $GVMIG$. (1)
- (iii) Find the number of different possible tours for Mark. (1)
- (iv) Write down the number of different possible tours for Mark if he were to visit n museums, starting and finishing at the Grand Hotel. (1)

(Total 14 marks)

Q16.

Salvadore is visiting six famous places in Barcelona: La Pedrera (L), Nou Camp (N), Olympic Village (O), Park Guell (P), Ramblas (R) and Sagrada Familia (S). Owing to the traffic system the time taken to travel between two places may vary according to the direction of travel.

The table shows the times, in minutes, that it will take to travel between the six places.

From \ To	La Pedrera (L)	Nou Camp (N)	Olympic Village (O)	Park Guell (P)	Ramblas (R)	Sagrada Familia (S)
La Pedrera (L)	—	35	30	30	37	35
Nou Camp (N)	25	—	20	21	25	40
Olympic Village (O)	15	40	—	25	30	29
Park Guell (P)	30	35	25	—	35	20
Ramblas (R)	20	30	17	25	—	25
Sagrada Familia (S)	25	35	29	20	30	—

- (a) Find the total travelling time for:
- (i) the route $LNOL$; (1)
- (ii) the route $LONL$. (1)
- (b) Give an example of a Hamiltonian cycle in the context of the above situation. (1)
- (c) Salvadore intends to travel from one place to another until he has visited all of the places before returning to his starting place.
- (i) Show that, using the nearest neighbour algorithm starting from Sagrada Familia (S), the total travelling time for Salvadore is 145 minutes. (3)
- (ii) Explain why your answer to part (c)(i) is an upper bound for the minimum travelling time for Salvadore. (2)
- (iii) Salvadore starts from Sagrada Familia (S) and then visits Ramblas (R). Given that he visits Nou Camp (N) before Park Guell (P), find an improved upper bound for the total travelling time for Salvadore. (3)

(Total 11 marks)

Q17.

- (a) Gill is solving a travelling salesperson problem.
- (i) She finds the following upper bounds: 7.5, 8, 7, 7.5, 8.5.
- Write down the best upper bound.

(1)

- (ii) She finds the following lower bounds: 6.5, 7, 6.5, 5, 7.

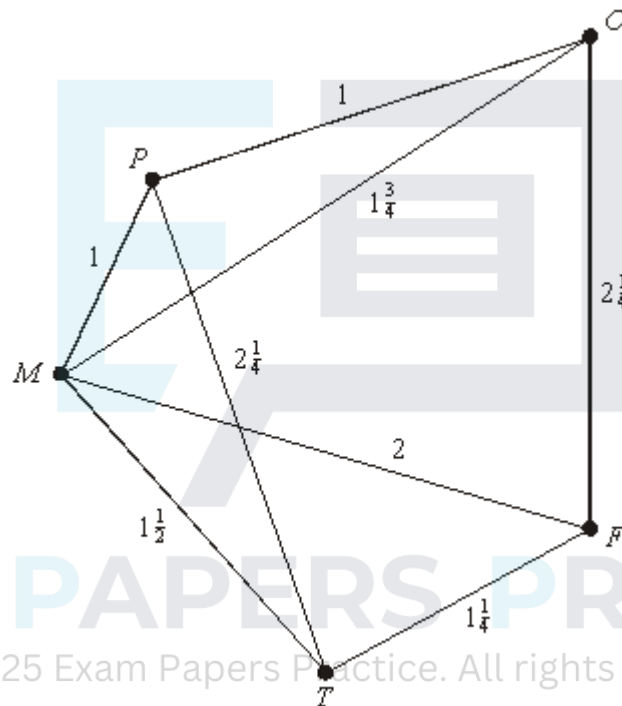
Write down the best lower bound.

(1)

- (b) George is travelling by plane to a number of cities. He is to start at F and visit each of the other cities at least once before returning to F .

The diagram in figure 1 shows the times of flights, in hours, between cities. Where no time is shown, there is no direct flight available.

Figure 1



- (i) Complete **Figure 2** to show the minimum times to travel between all pairs of cities.

Figure 2

	M	P	O	T	F
M	–	1	$1\frac{3}{4}$	$1\frac{1}{2}$	2
P	1	–	1	$2\frac{1}{4}$	
O	$1\frac{3}{4}$	1	–		$2\frac{1}{4}$
T	$1\frac{1}{2}$	$2\frac{1}{4}$		–	$1\frac{1}{4}$
F	2		$2\frac{1}{4}$	$1\frac{1}{4}$	–

(2)

- (ii) Find an upper bound for the minimum total flying time by using the route $FTPOMF$.

(1)

- (iii) Using the nearest neighbour algorithm starting from F , find an upper bound for the minimum total flying time.

(4)

- (iv) By deleting F , find a lower bound for the minimum total flying time.

(5)

(Total 14 marks)

Q18.

Rob delivers bread to six shops A , B , C , D , E and F . Each day, Rob starts at shop A , travels to each of the other shops before returning to shop A . The table shows the distances, in miles, between the shops.

	A	B	C	D	E	F
A	–	8	6	9	12	7
B	8	–	10	14	13	8
C	6	10	–	7	16	10
D	9	14	7	–	15	5
E	12	13	16	15	–	11

F	7	8	10	5	11	–
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- (a) (i) Find the length of the tour $ABCDEF A$. (2)
- (ii) Find the length of the tour obtained by using the nearest neighbour algorithm starting from A . (4)
- (b) By deleting A , find a lower bound for the length of a minimum tour. (5)
- (c) By deleting F , another lower bound of 45 miles is obtained for the length of a minimum tour.
- The length of a minimum tour is T miles. Write down the smallest interval for T which can be obtained from your answers to parts (a) and (b), and the information given in this part. (3)
- (Total 14 marks)**

Q19.

Mia is on holiday in Venice. There are five places she wishes to visit: Rialto (R), St Mark's (S), Murano (M), Burano (B) and Lido (L). Boat services connect the five places. The table shows the times, in minutes, to travel between the places.

Mia wishes to keep her travelling time to a minimum.

	Rialto (R)	St Mark's (S)	Murano (M)	Burano (B)	Lido (L)
Rialto (R)	–	15	55	75	25
St Mark's (S)	15	–	90	60	20
Murano (M)	55	90	–	25	80
Burano (B)	75	60	25	–	50
Lido (L)	25	20	80	50	–

- (a) (i) Find the length of the tour $SRMBLS$. (2)
- (ii) Find the length of the tour using the nearest neighbour algorithm starting from S . (4)
- (b) By deleting Burano (B), find a lower bound for the length of the minimum tour. (5)

- (c) Sketch a network showing the edges that give the lower bound found in part (b) and comment on its significance.

(2)

(Total 13 marks)

Q20.

David is going to buy a holiday home in one of five villages: Aragona (A), Belvedere (B), Comitini (C), Grotte (G) and St Vincenzo (V). David is going to visit the five villages. He intends to travel from one village to the next until he has visited all of the villages, before returning to his starting village.

The distances, in kilometres, between the villages are shown in the following table.

	Aragona (A)	Belvedere (B)	Comitini (C)	Grotte (G)	St Vincenzo (V)
Aragona (A)	—	11	10	13	8
Belvedere (B)	11	—	13	11	9
Comitini (C)	10	13	—	5	4
Grotte (G)	13	11	5	—	6
St Vincenzo (V)	8	9	4	6	—

- (a) (i) Find the length of the tour *AGCVBA*.

(1)

- (ii) Find the length of the tour obtained by using the nearest neighbour algorithm starting from *A*.

(4)

- (b) (i) By deleting *A*, find a lower bound for the length of a minimum tour.

(3)

- (ii) By deleting *G*, find another lower bound for the length of a minimum tour.

(3)

- (c) The length of a minimum tour is *T* kilometres.

Write down the smallest interval for *T* which can be obtained from your answers to parts (a) and (b).

(2)

(Total 13 marks)

Q21.

Renata is visiting a theme park. There are five film sets: Alien (A), Backdraft (B), Conan (C), Die Hard (D) and E.T. (E). The table shows the times, in minutes, taken by Renata to walk from one film set to another.

Renata wants to visit all of the five film sets starting and finishing at A.

	Alien (A)	Backdraft (B)	Conan (C)	Die Hard (D)	E.T. (E)
Alien (A)	–	7	11	9	3
Backdraft (B)	7	–	8	6	12
Conan (C)	11	8	–	5	4
Die Hard (D)	9	6	5	–	10
E.T. (E)	3	12	4	10	–

- (a) Renata decides to visit the film sets in the order ABCDEA. Find the walking time of this tour.

(1)

- (b) Explain why this answer can be considered as being an upper bound for the minimum walking time of Renata's tour.

(2)

- (c) Use the nearest neighbour algorithm, starting from A, to find another upper bound.

(4)

- (d) By deleting A, find a lower bound for Renata's tour.

(4)

- (e) Renata thinks that she can reduce her walking time to 20 minutes. Explain why this is impossible.

(1)

(Total 12 marks)

Q22.

Tony is going on a touring holiday in America. The following table shows the five cities that he is to visit. The figures represent the distances, in miles, between the cities.

	Los Angeles (LA)	Las Vegas (LV)	Palm Springs (PS)	Santa Barbara (SB)	San Diego (SD)

Los Angeles (LA)	—	190	210	90	185
Las Vegas (LV)	190	—	180	140	300
Palm Springs (PS)	210	180	—	230	150
Santa Barbara (SB)	90	140	230	—	250
San Diego (SD)	185	300	150	250	—

Tony is to start his tour at Los Angeles, visiting each city once before returning to Los Angeles.

- (a) Use the nearest neighbour algorithm, starting from Los Angeles, to find an upper bound for a tour of the five cities. (4)
- (b) By deleting Los Angeles, obtain a lower bound for a tour of the five cities. (5)
- (c) An optimal tour has length T miles. Using your answers to parts (a) and (b), write down a statement about T . (1)

(Total 10 marks)

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Q23.

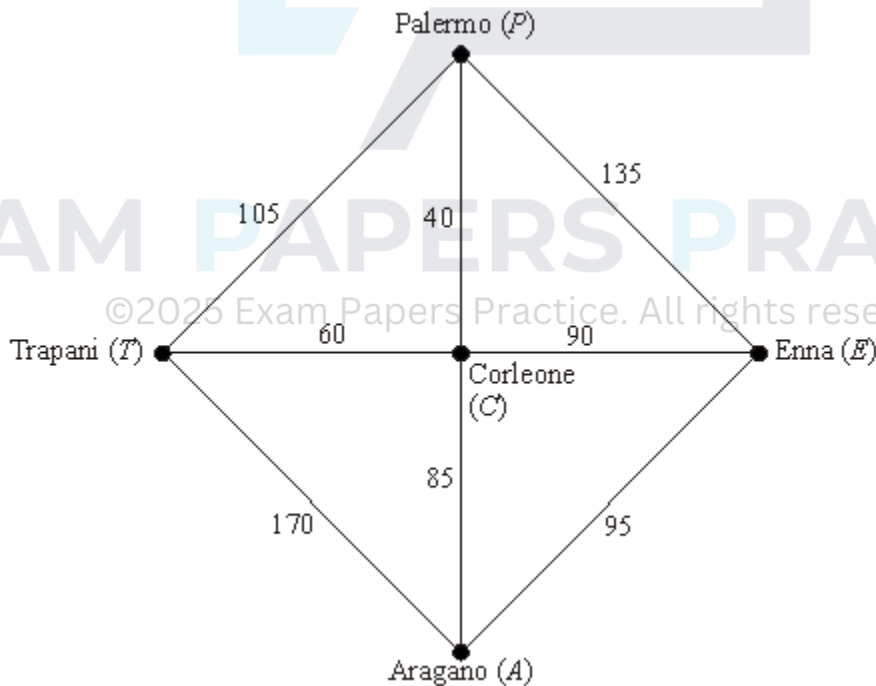
Peppi is a mobile hairdresser based at Chadderton (C). His day's appointments are at six places: Ashton (A), Bardsley (B), Chadderton (C), Derker (D), Edenfield (E) and Failsworth (F). He intends to travel from one place to the next until he has visited all of the places, starting and finishing at Chadderton. The following table shows the times, in minutes, taken to travel between the six places.

	Ashton (A)	Bardsley (B)	Chadderton (C)	Derker (D)	Edenfield (E)	Failsworth (F)
Ashton (A)	—	18	14	17	31	15
Bardsley (B)	18	—	16	22	27	18
Chadderton (C)	14	16	—	13	22	15
Derker (D)	17	22	13	—	29	18
Edenfield (E)	31	27	22	29	—	30
Failsworth (F)	15	18	15	18	30	—

- (a) (i) Use the nearest neighbour algorithm, starting and finishing at Chadderton, to find a possible route for Peppi and the total travelling time it would take. (5)
- (ii) Explain why your answer to part (a) (i) gives an upper bound for Peppi's minimum possible total travelling time. (2)
- (b) By deleting Chadderton, find a lower bound for the minimum total travelling time. (5)
- (c) Given that the minimum total travelling time is T minutes, use your answers to parts (a) and (b) to write down an interval within which T must lie. (1)
- (Total 13 marks)**

Q24.

A Sicilian market trader, Vito, who lives in Corleone (C), has to visit the towns Palermo (P), Aragona (A), Enna (E) and Trapani (T) on a particular day before returning to Corleone. He is trying to find the route that will minimise his travelling distance. The following diagram shows the distances, in kilometres, between the towns.



- (a) Complete the table below, in which the entries are the shortest distances, in kilometres, between pairs of vertices.

	Aragona (A)	Corleone (C)	Enna (E)	Palermo (P)	Trapani (T)
Aragona (A)	–	85	95	125	145
Corleone	85	–	90	40	60

(C)					
Enna (E)	95	90	–		
Palmero (P)	125	40		–	
Trapani (T)	145	60			–

(3)

- (b) Use the nearest neighbour algorithm on the table above to find an upper bound for the minimum length of a tour of this network that starts and finishes at Corleone.

(4)

- (c) Vito decides to follow the route given by the nearest neighbour algorithm. Write down his route.

(2)

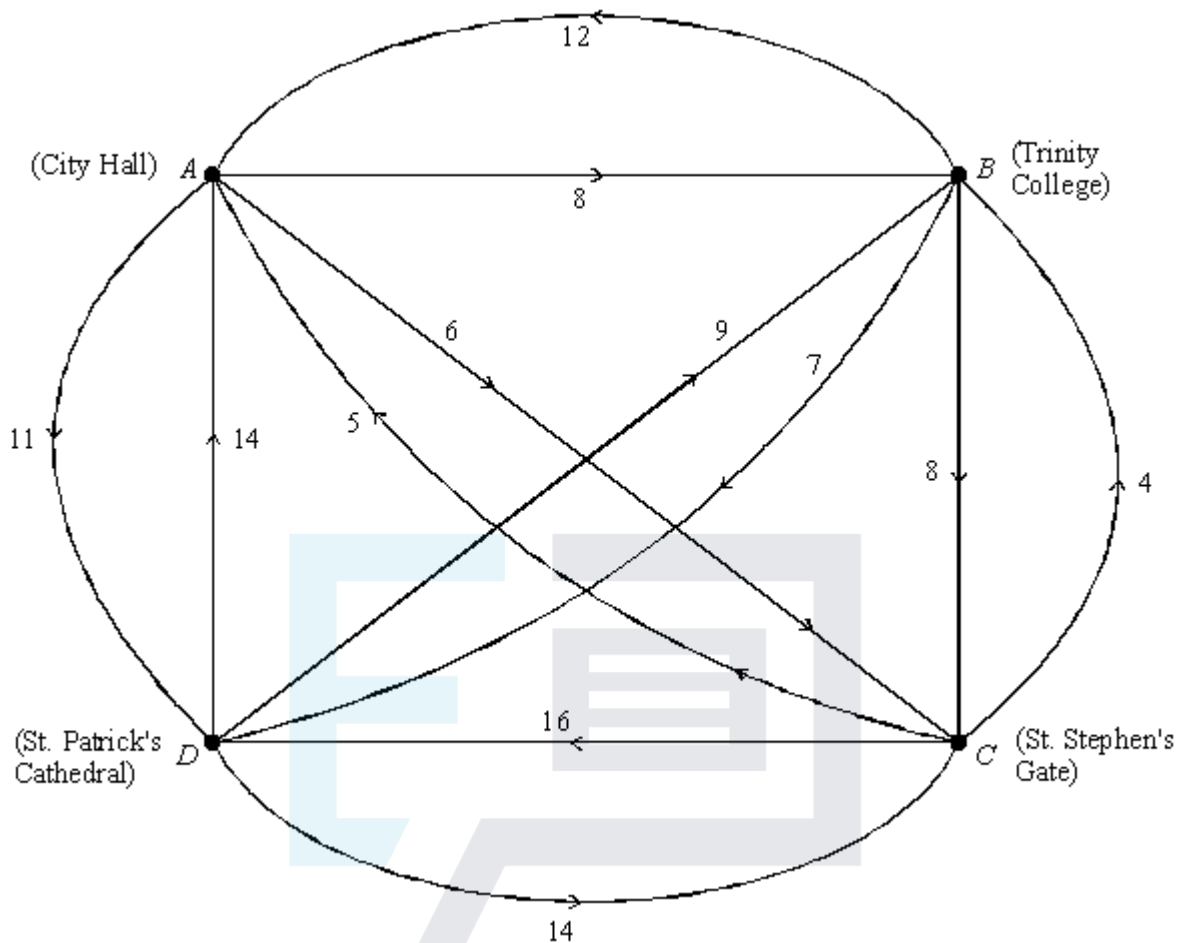
(Total 9 marks)

Q25.

A family is staying in Dublin and is to visit four places of interest in the city. There is a one-way traffic system in the city and consequently the time taken to travel from A to B is different from the time taken to travel from B to A .

The following diagram shows the four places to be visited together with the time taken, in minutes, to travel between each two places.

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- (a) The family starts the tour at A and visits each of the other three places once before returning to A.

- (i) Use the nearest neighbour algorithm to find an upper bound for the travelling time of the tour.

(4)

- (ii) Find the number of different possible tours.

(2)

- (b) Write down an expression for the number of possible tours if the family were to start at A and visit each of n other places once before returning to A.

(2)

(Total 8 marks)

Q26.

A machine is used for producing sweets in six flavours. The machine produces one flavour of sweet at a time. It needs to be cleaned before changing flavours. The time taken to clean the machine depend on the two flavours involved and these times, in minutes, are given in the table below.

The machine is to be set to produce each flavour in sequence before repeating the cycle.

The machine can only start with Raspberry (R) or Strawberry (S).

		To					
		Blackcurrant (B)	Lime (L)	Orange (O)	Plum (P)	Raspberry (R)	Strawberry (S)
From	Blackcurrant (B)	–	25	20	20	27	25
	Lime (L)	15	–	10	11	15	30
	Orange (O)	5	30	–	15	20	19
	Plum (P)	20	25	15	–	25	10
	Raspberry (R)	10	20	7	15	–	15
	Strawberry (S)	15	25	19	10	20	–

- (a) (i) Show that, using the nearest neighbour algorithm starting with Strawberry (S), the total cleaning time for one cycle is 85 minutes.

(3)

- (ii) Use the same method starting with Raspberry (R) to find the total cleaning time for one cycle, which is less than 85 minutes.

(4)

- (b) Explain why each of your answers to part (a) is an upper bound for the minimum **total** cleaning time for one cycle.

(2)

- (c) Given that the machine produces S first followed by R, find an improved upper bound for the minimum total cleaning time for one cycle.

(5)

(Total 14 marks)

Q27.

A message is to be taken by a secretary to each of five classrooms. The secretary is based in the school office and will go to each classroom once before returning to the school office.

The table gives the times, in seconds, taken by the secretary to walk between pairs of rooms.

	Office	Room 1	Room 2	Room 3	Room 4	Room 5
Office	—	44	52	54	50	48
Room 1	44	—	56	50	54	53
Room 2	52	56	—	66	62	61

Room 3	54	50	66	—	64	62
Room 4	50	54	62	64	—	60
Room 5	48	53	61	62	60	—

- (a) (i) The secretary leaves the office and visits rooms 1, 2, 3, 4 and 5 in that order before returning to the office.

Find the walking time for this tour.

(2)

- (ii) Explain why this answer may be considered as an upper bound for the minimum total walking time for the secretary's tour.

(2)

- (b) Use the nearest neighbour algorithm, starting from the office, to obtain an improved upper bound.

(4)

- (c) By initially ignoring the office, find a lower bound for the walking time of the secretary's tour.

(5)

(Total 13 marks)

Q28.

Roger, a football supporter, is to visit each of six football grounds. He decides to travel from one ground to the next until he has visited all of the grounds, starting and finishing at Man. City. The following table shows the distances, in miles, between the grounds.

	Man. City	Burnley	Crewe	Preston	Stockport	Tranmere
Man. City	-	19	26	32	8	31
Burnley	19	-	43	21	22	36
Crewe	26	43	-	42	19	23
Preston	32	21	42	-	36	26
Stockport	8	22	19	36	-	27
Tranmere	31	36	23	26	27	-

- (a) Use the nearest neighbour algorithm, starting and finishing at Man. City, to find an upper bound for the total distance Roger must travel.

(4)

- (b) By initially ignoring Man. City, find a lower bound for the total distance he must travel in visiting the six grounds.

(5)

- (c) Using your answer to parts (a) and (b), write down inequalities for D , the total

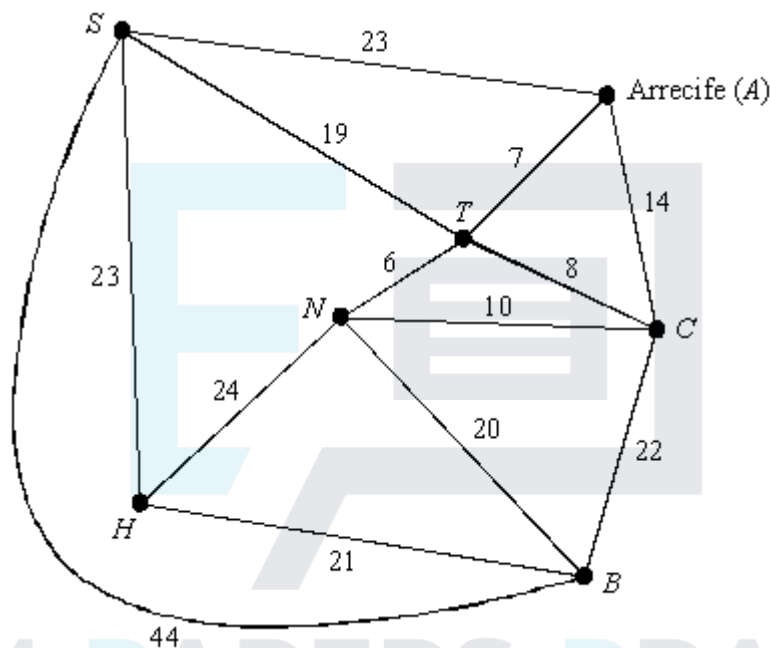
distance, in miles, that Roger has to travel.

(1)

(Total 10 marks)

Q29.

The island of Lanzarote is short of water and the local government is to implement a new system so that water is distributed to all the main towns on the island. The main source of water is at Arrecife (A). The diagram shows the main towns and the distances, in kilometres, between them.



- (a) Using Prim's algorithm starting from A, and showing your working at each stage, find the minimum length of piping needed. Show your final network of piping as a minimum spanning tree.

(5)

- (b) A tourist lands at the airport at Arrecife (A) and is to tour the island, visiting each of the other six towns before returning to A.

- (i) By deleting A, find a lower bound for the length of such a tour.

(3)

- (ii) Explain why a tour of such a length is impossible in this situation.

(2)

(Total 10 marks)

Q30.

It is the year 2302 and space travel between the planets of the solar system has become commonplace.

Sarah, who is based on Mars, is planning a tour of four other planets before returning to Mars. The planets orbit the sun, and the distance between each pair of planets varies. The following table gives the distance between planets, in space units, in terms of a variable x .

	Mars (M)	Jupiter (J)	Saturn (S)	Neptune (N)	Pluto (P)
Mars (M)	-	$3x + 6$	$x + 24$	$2x + 17$	$5x$
Jupiter (J)	$3x + 6$	-	$x + 4$	$2x - 3$	$3x - 2$
Saturn (S)	$x + 24$	$x + 4$	-	$x + 13$	$2x - 1$
Neptune (N)	$2x + 17$	$2x - 3$	$x + 13$	-	$x + 6$
Pluto (P)	$5x$	$3x - 2$	$2x - 1$	$x + 6$	-

- (a) Given that Jupiter is the nearest planet to Mars, find three inequalities in x .

(3)

Sarah plans a tour using the nearest neighbour algorithm and finds that, for one particular **integer** value of x , this algorithm gives a **unique** tour of $M J S P N M$.

- (b) By considering the fact that Sarah's tour visits Saturn immediately after Jupiter, find two further inequalities in x .

(3)

- (c) Hence find the value of the integer x , and the total length of Sarah's tour.

(5)

(Total 11 marks)

Q31.

[Figures 1 and 2, printed below, are provided for use in answering parts (a) and (b) of this question.]

	Rochdale	Castleton	Middleton	Shaw	Milnrow	Littleborough	Whitworth
Rochdale	-	3	7	8	6	5	4
Castleton	3	-	9	6	8	7.5	6.5
Middleton	7	9	-	14	12	11.5	12
Shaw	8	6	14	-	13	12	11
Milnrow	6	8	12	13	-	10	9
Littleborough	5	7.5	11.5	12	10	-	8
Whitworth	4	6.5	12	11	9	8	-

Figure 1

	Rochdale	Castleton	Middleton	Shaw	Milnrow	Littleborough	Whitworth
Rochdale	–	3	7	8	6	5	4
Castleton	3	–	9	6	8	7.5	6.5
Middleton	7	9	–	14	12	11.5	12
Shaw	8	6	14	–	13	12	11
Milnrow	6	8	12	13	–	10	9
Littleborough	5	7.5	11.5	12	10	–	8
Whitworth	4	6.5	12	11	9	8	–

Figure 2

A company based in Rochdale produces a free newspaper for distribution locally. The following table shows the six surrounding towns that receive the free paper. The figures represent the time, in minutes, to travel between the towns. The company delivery van has to travel from Rochdale to each one of the other towns, before returning to Rochdale.

	Rochdale	Castleton	Middleton	Shaw	Milnrow	Littleborough	Whitworth
Rochdale	–	3	7	8	6	5	4
Castleton	3	–	9	6	8	7.5	6.5
Middleton	7	9	–	14	12	11.5	12
Shaw	8	6	14	–	13	12	11
Milnrow	6	8	12	13	–	10	9
Littleborough	5	7.5	11.5	12	10	–	8
Whitworth	4	6.5	12	11	9	8	–

- (a) Find a minimum connector for the seven towns, stating its length. (5)
- (b) Use the nearest neighbour algorithm, starting from Rochdale, to find an upper bound for a tour of the seven towns. (4)
- (c) By deleting Rochdale from the minimum connector found in part (a), obtain a lower bound l for a tour of the seven towns.

(4)
(Total 13 marks)

Q32.

A sweet company has a production line making batches of seven different flavours of sweets A, B, C, D, E, F, G . The changeover times, in minutes, from the production line being set up for one flavour to it being set up for another flavour are given in the following table.

	A	B	C	D	E	F	G
A	–	13	17	18	16	15	14
B	13	–	19	16	18	17.5	16.5
C	17	19	–	24	22	21.5	22
D	18	16	24	–	23	22	21
E	16	18	22	23	–	20	19
F	15	17.5	21.5	22	20	–	18
G	14	16.5	22	21	19	18	–

(a) Normally sweets are produced in the order A, B, C, D, E, F, G . The production line is then set up to start with flavour A the next day.

(i) Find the total time taken up by the changeovers.

(2)

(ii) Explain why this answer can be considered to be an upper bound for this travelling salesman problem.

(2)

(b) Use the nearest neighbour algorithm, starting at A , to find a reduced time spent on changeovers.

(4)

(c) By initially ignoring sweet A , find a lower bound for the changeover times.

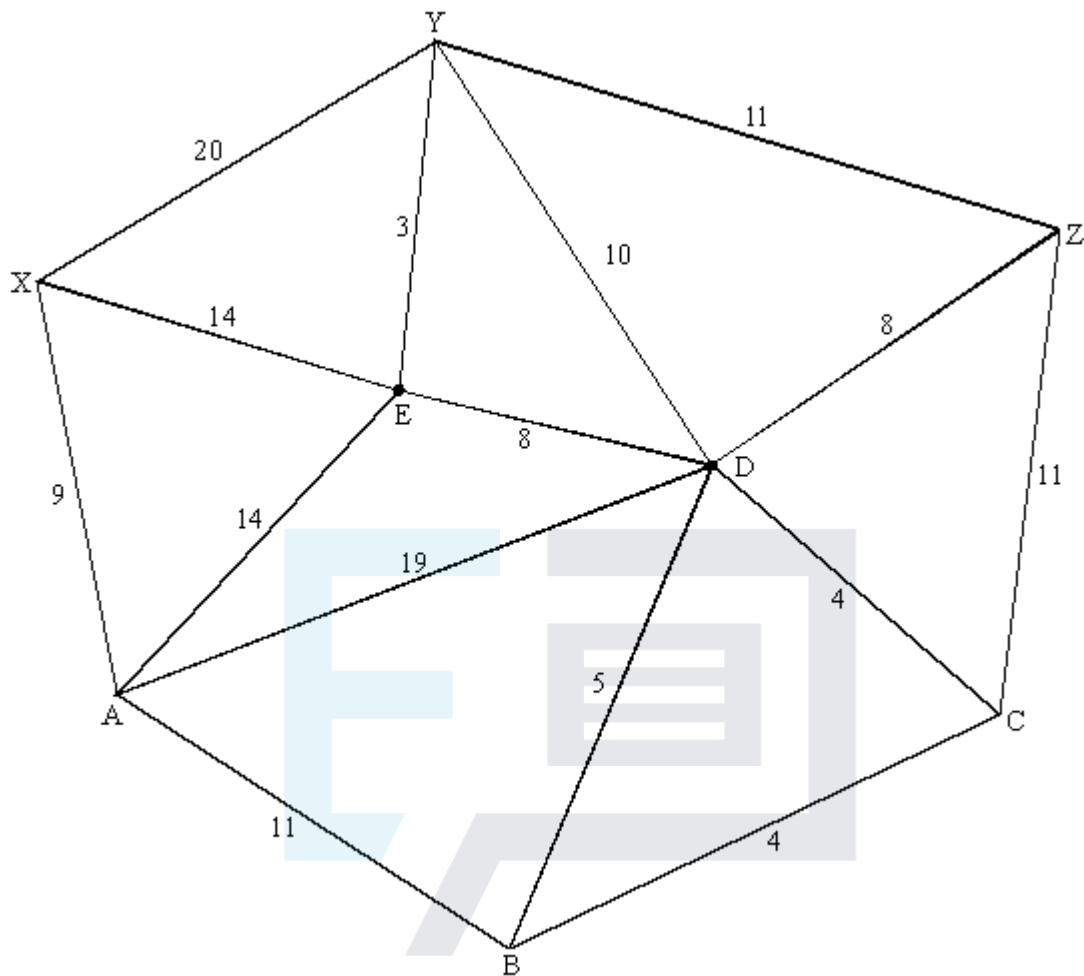
(5)

(Total 13 marks)

Q33.

A travelling salesman wishes to visit each of eight towns, returning to the town where he started.

The following diagram shows the distances, in miles, between the eight towns.



- (a) Find a minimum spanning tree for the eight towns, stating its length. (3)
- (b) Use the nearest neighbour algorithm to find an improved upper bound starting
- (i) from town Z , (2)
- (ii) from town Y . (3)
- (c) By deleting town X obtain a lower bound for a tour. (3)
- (d) Hence write down inequalities for L , the minimum length of a tour of the eight towns. (1)

(Total 12 marks)

Q34.

A saleswoman has to visit six towns during day before returning home. The distances, in miles, between the six towns is given in the following table.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	–	22	26	16	30	26
<i>B</i>	22	–	20	32	10	12
<i>C</i>	26	20	–	34	16	16
<i>D</i>	16	32	34	–	34	32
<i>E</i>	30	10	16	34	–	8
<i>F</i>	26	12	16	32	8	–

- (a) Find a minimum spanning tree for the six towns.

(4)

To try to find a shorter tour, the saleswoman deletes town *E* and obtains a lower bound of 84 miles. She obtains the same result by deleting town *F*.

- (b) By deleting the other four towns in turn, find four further lower bounds for the distance she must travel in visiting the six towns

(6)

(Total 10 marks)

Q35.

During a general election campaign a politician, who is based at *A*, has to visit towns *B*, *C* and *D* on a particular day before returning to *A*. He is trying to find the tour that will minimise his travelling distance. The following table shows the distances, in kilometres, between the four towns.

To From	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	–	120	80	60
<i>B</i>	120	–	75	90
<i>C</i>	80	75	–	65
<i>D</i>	60	90	65	–

- (a) Use the nearest unvisited town to find a possible tour and state its length.

(3)

- (b) Explain why the length of this tour may be considered to be an upper bound.

(1)

(Total 4 marks)

Q36.

Paul, who lives in Manchester, wide visit one of the universities in each of Manchester, Liverpool, Sheffield and Birmingham before he fills in his university application form. He decides to spend part of his holiday visiting each university by travelling directly from one to the next. The table below shows the distances in kilometres between the universities.

	Manchester	Liverpool	Sheffield	Birmingham
Manchester	—	56	61	129
Liverpool	56	—	116	150
Sheffield	61	116	—	122
Birmingham	129	150	122	—

By deleting each university in turn, find four lower bounds and hence the best lower bound for the distance which Paul must travel between the four universities.

(Total 11 marks)

Q37.

A machine is used for producing ice cream in six flavours. The machine produces one flavour of ice cream at a time. It needs to be cleaned before changing flavours. The times taken to clean the machine depend on the two flavours involved and these times, in minutes, are given in the table. The machine is set to produce each flavour in sequence before repeating the cycle. The machine can start the cycle with any flavour.

To From	Vanilla	Lemon	Orange	Raspberry	Coffee	Mint
Vanilla	—	40	35	35	42	40
Lemon	30	—	25	26	30	45
Orange	20	45	—	30	35	34
Raspberry	35	40	30	—	40	25
Coffee	25	35	22	30	—	30
Mint	30	40	34	25	35	—

Upper bound times in excess of three hours are produced by using the nearest neighbour algorithm when starting with Vanilla or Raspberry. Use the same method to find **four** further upper bounds for the total cleaning time, showing that only one produces a total time of under 175 minutes.

(Total 6 marks)

Q38.

The towns P , Q , R , S , T , U , V and W are all linked to each other by direct roads whose lengths, in miles, are shown in the following table:

	<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>
<i>P</i>	–	5	4	3	4	5	4	6
<i>Q</i>	5	–	7	5	3	5	8	8
<i>R</i>	4	7	–	6	8	6	3	5
<i>S</i>	3	5	6	–	4	6	9	6
<i>T</i>	4	3	8	4	–	7	8	6
<i>U</i>	5	5	6	6	7	–	8	3
<i>V</i>	4	8	3	9	8	8	–	8
<i>W</i>	6	8	5	6	6	3	8	–

(a) An estate agent wants to start at the town *P* and visit each of the other towns once before returning to *P*.

(i) Use the nearest neighbour algorithm to find one possible route for the estate agent and state its total length.

(4)

(ii) On the table below, ignore the town *P* and find the length of a minimum connector of the towns *Q*, *R*, *S*, *T*, *U*, *V* and *W*.

	<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>
<i>P</i>	–	5	4	3	4	5	4	6
<i>Q</i>	5	–	7	5	3	5	8	8
<i>R</i>	4	7	–	6	8	6	3	5
<i>S</i>	3	5	6	–	4	6	9	6
<i>T</i>	4	3	8	4	–	7	8	6
<i>U</i>	5	5	6	6	7	–	8	3
<i>V</i>	4	8	3	9	8	8	–	8
<i>W</i>	6	8	5	6	6	3	8	–

(4)

(iii) Deduce a lower bound for the estate agent's route and comment on its significance to your answer to part (a)(i).

(3)

- (b) The estate agent would then like to look at the houses on each of the roads joining the towns. The estate agent starts at P , covers each of the roads at least once, and finishes at P .

- (i) Explain why it will be necessary to cover at least four of the roads more than once.

(2)

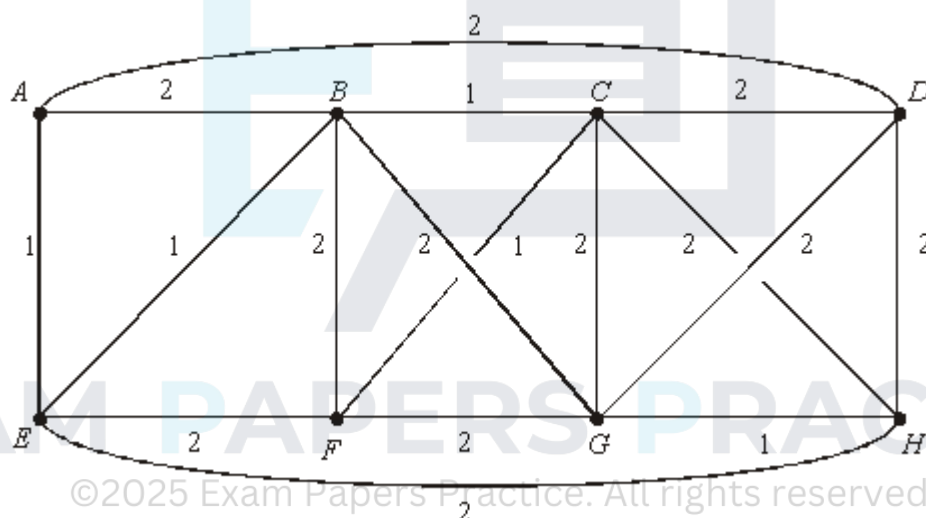
- (ii) Given that the total length of all the roads is 160 miles, explain how you know that all the roads can be covered with a round trip of 172 miles.

(3)

(Total 16 marks)

Q39.

The network below shows the phone lines connecting eight offices $A-H$. The cost per minute for a phone call between two offices is either 1p or 2p as shown on the network.



- (a) For the transfer of information, the offices wish to keep a cycle of eight phone lines open. In this way one office will be linked to another, which will be linked to another, and so on until the eighth one is then linked back to the first.

- (i) Use a nearest neighbour approach, starting at A , to find one suitable cycle of phone lines to keep open.

(3)

- (ii) Explain why the cycle which you found in part (i) is in fact the cheapest possible.

(1)

- (b) The phone company wishes to check all the lines shown in the above network by starting at A , passing a message along one line, and then the receiver of the message will pass it along another line, and so on until the message is eventually received back at A , having used each of the lines at least once.

- (i) Use the Chinese postperson algorithm to find which lines will be used twice when this checking is done in the most economical way.

(4)

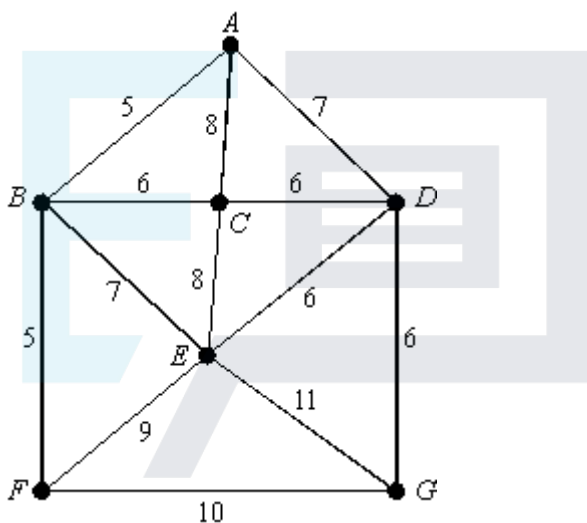
- (ii) The condition that the message ends back at A is dropped, and the company merely requires a route which uses each line at least once. In the new most economical route, which line, or lines, will be used more than once, and at which office will the message end?

(2)

(Total 10 marks)

Q40.

The numbers of parking meters on the roads in a town centre are shown in the following network.



- (a) A traffic warden wants to start at A , walk along the roads passing each meter at least once, and finish back at A . She wishes to choose her route in order to minimise the number of meters that she passes more than once.
- (i) Explain how you know that it will be necessary for her to pass some meters more than once.
- (ii) Apply the Chinese postperson algorithm to find the minimum number of meters which she will have to pass more than once, and give an example of a suitable route.
- (b) At each of the junctions A, B, C, D, E, F and G there is a set of traffic lights. The traffic warden is asked to make a journey, starting and finishing at A , to check that each set of traffic lights is working correctly. Find a suitable route for her which passes 50 or fewer meters.

(1)

(6)

(4)

(Total 11 marks)

Q41.

In the following table $A - F$ represent tourist attractions in an alpine village, and when there is a direct path between them its length in metres is given as an entry in the table.

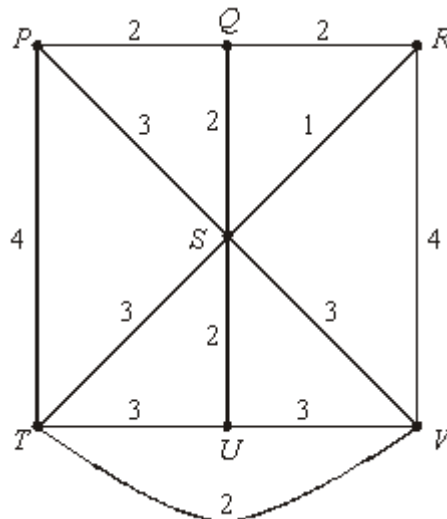
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	–	–	100	110	150	150
<i>B</i>	–	–	180	170	180	160
<i>C</i>	100	180	–	120	–	–
<i>D</i>	110	170	120	–	150	140
<i>E</i>	150	180	–	150	–	120
<i>F</i>	150	160	–	140	120	–

- (a) After each fall of snow the local tourist board wishes to clear some paths so that it is possible to walk between any two of the attractions on snow-free paths. By applying an appropriate algorithm, find the minimum total length of paths which must be cleared, and state which paths must be cleared in order to achieve this minimum. (5)
- (b) Illustrate as a graph with six vertices *A* – *F* the minimum connector which you found in part (a). (2)
- (c) The tourist board decides to clear two further paths so that it is possible to do a round tour visiting all the attractions on snow-free paths. Find **two** additional paths which should be cleared in order to achieve this and find the length of the round tour. (3)

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Q42.

The network **N** is illustrated below:



- (a) Use the Chinese postperson algorithm to find the length of a shortest closed walk in **N** which uses all its edges.

(5)

- (b) Apply the nearest neighbour algorithm, starting and finishing at S , to construct a Hamiltonian cycle of $\mathbf{N}$. State its length.

(In this case the algorithm works even though the network is not complete.)

(4)

- (c) (i) By considering the vertex U , explain why every Hamiltonian cycle of $\mathbf{N}$ uses at least one edge of length 3.
- (ii) By considering $\mathbf{N}$ with edges PT and RV deleted, explain briefly why each Hamiltonian cycle of $\mathbf{N}$ uses at least one edge of length 4.
- (iii) Show that the Hamiltonian cycle which you constructed in part (b) is the shortest possible in $\mathbf{N}$.

(4)

- (d) The network $\mathbf{N}$ represents seven computer centres P, Q, R, S, T, U, V and the direct cable links between them. The number on each edge shows the time, in seconds, that it takes for a piece of information to pass along that link. Whenever a piece of information is received at one centre and then passed on to another there is a further 20 second delay.

- (i) A piece of information has to be sent from S to another centre and then passed on to another and then passed on again, until it is eventually sent back to S having been received at least once at each of the other computer centres. Use your earlier answers to find the minimum time that this will take.

(2)

- (ii) A technician based at S wants to check every link in the system by sending a piece of information from S to another centre and then on to another and then on again, until it is eventually sent back to S having used **each link** at least once. Use your earlier answers to find the minimum time that this will take.

(3)

(Total 18 marks)

Q43.

A country policeman wants to start at the police station P and visit each of the villages Q, R, S, T, U, V once before returning to the station. The distances (in miles) by the easiest routes are shown in the table:

	P	Q	R	S	T	U	V
P	–	5	4	3	4	5	4
Q	5	–	7	3	4	5	8
R	4	7	–	6	8	6	7
S	3	3	6	–	4	6	9

<i>T</i>	4	4	8	4	–	7	8
<i>U</i>	5	5	6	6	7	–	8
<i>V</i>	4	8	7	9	8	8	–

(a) Use the nearest neighbour algorithm to find one possible route for the policeman. (4)

(b) Find the length of a minimum connector of the villages *Q*, *R*, *S*, *T*, *U*, *V*. (3)

(c) Deduce that the length L of the shortest route for the policeman satisfies

$$32 \leq L \leq 34.$$

(4)

(Total 11 marks)

Q44.

A researcher starts at the desk, *D*, of the British Library and wishes to consult books at locations *E*, *F*, *G*, *H*, *I* and *J* in the library. The time, in minutes, needed to walk between any two places is given in the table.

	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
<i>D</i>	–	7	6	9	7	10	8
<i>E</i>	7	–	11	6	12	10	7
<i>F</i>	6	11	–	10	11	11	12
<i>G</i>	9	6	10	–	13	11	8
<i>H</i>	7	12	11	13	–	8	12
<i>I</i>	10	10	11	11	8	–	9
<i>J</i>	8	7	12	8	12	9	–

(a) Use the nearest neighbour algorithm, starting at *D*, to find one appropriate order in which the researcher might visit the six locations before returning to *D*. State the total time needed for walking around this particular route.

	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
<i>D</i>	–	7	6	9	7	10	8
<i>E</i>	7	–	11	6	12	10	7
<i>F</i>	6	11	–	10	11	11	12
<i>G</i>	9	6	10	–	13	11	8
<i>H</i>	7	12	11	13	–	8	12
<i>I</i>	10	10	11	11	8	–	9
<i>J</i>	8	7	12	8	12	9	–

Figure 1

(4)

- (b) Find a minimum connector of just the six locations *E*, *F*, *G*, *H*, *I* and *J*, and state its total length. Draw a tree representing your minimum connector.

	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>
<i>D</i>	–	7	6	9	7	10	8
<i>E</i>	7	–	11	6	12	10	7
<i>F</i>	6	11	–	10	11	11	12
<i>G</i>	9	6	10	–	13	11	8
<i>H</i>	7	12	11	13	–	8	12
<i>I</i>	10	10	11	11	8	–	9
<i>J</i>	8	7	12	8	12	9	–

Figure 2

(5)

- (c) Explain why, in this case, the route which you found in part (a) is definitely the shortest possible.

(3)

(Total 12 marks)

Q45.

The following matrix gives the distances in miles between each of five locations.

		To				
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
From	<i>A</i>	–	45	60	50	90
	<i>B</i>	45	–	67	25	80
	<i>C</i>	60	67	–	50	70
	<i>D</i>	50	25	50	–	35
	<i>E</i>	90	80	70	35	–

- (a) Find the minimum length of cable which would be needed to connect these locations together, given that a connection between two locations may be made via other locations.

(4)

- (b) A tour is a journey which starts from a location, visits each other location once and only once, and returns to the starting location. Use an appropriate method to find a tour, starting and finishing at *A*, with a total length of less than 270 miles.

(4)

- (c) By temporarily deleting *E* and its connections, and by finding a minimum connector for the remaining network, construct a lower bound for the minimum tour length.

(4)

- (d) Give an upper bound for the minimum tour length, and explain why it is an upper bound.

(2)

(Total 14 marks)